Agenda

* Assumptions of Linear Regression
  - Assumption of Linearity
  - No multicolinearity - VIF (variance Inflation Factor)
  - Normality of Residuals
  - no Heteroskedasticity
  - No Autocorrelation

- Assignments

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Recap

* Implementation

$$R^2_score = 1 - \frac{SS_{res}}{SS_{total}}$$

** Adjusted R^2_score = $1 - \left[\frac{(1-R^2)(n-1)}{(n-d-1)}\right]$ **

R^2 ↑ R-Squared

Real world → [0, 1]

Theoritically (−∞, 1]

* Intro to stats model → OLS → ordinary least squares

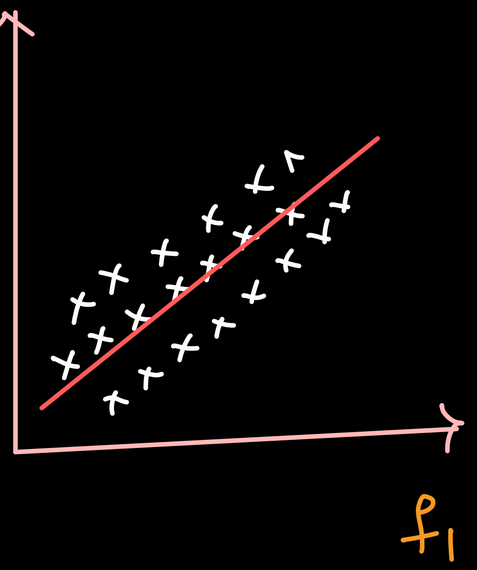
Assumption of Lin Reg

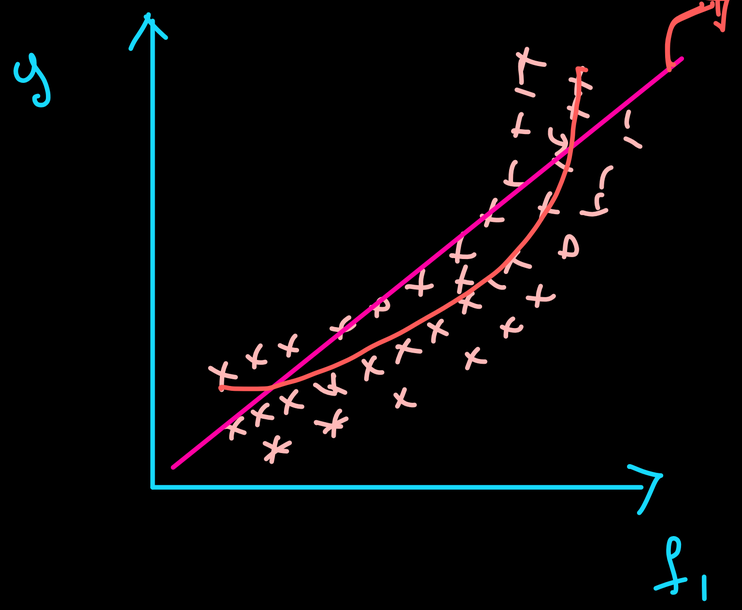
1. Assumption of Linearity
2. No multicolinearity
3. Normality of Residuals
4. No Heteroskedasticity
5. No Autocorrelation

(1) **Linearity** ✓✓ : There should be linear relationship b/w features and target

Eg:

(1)

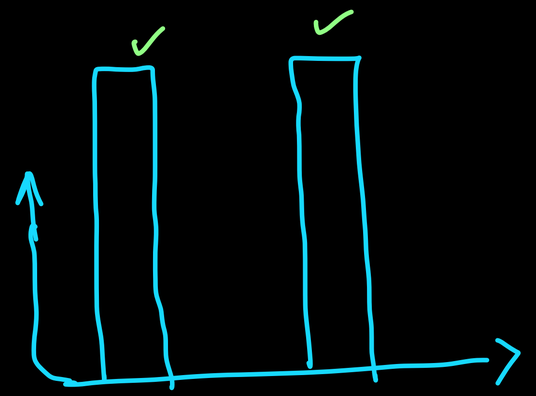




✓✓

R2 = 60-70%

(2) No multicolinearity :



what is colinearity ?

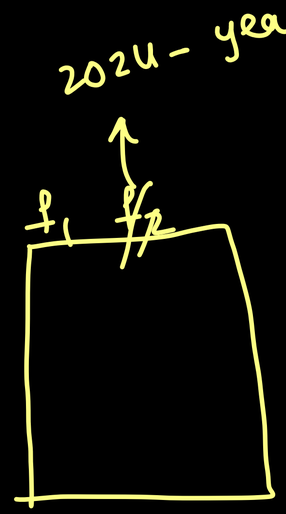
↳ collinearity

Eg: $f_1, f_2 \Rightarrow$ $f_2 = \alpha_1 f_1 + \beta_0$

age, year

2015

age = 2024 − year = 9

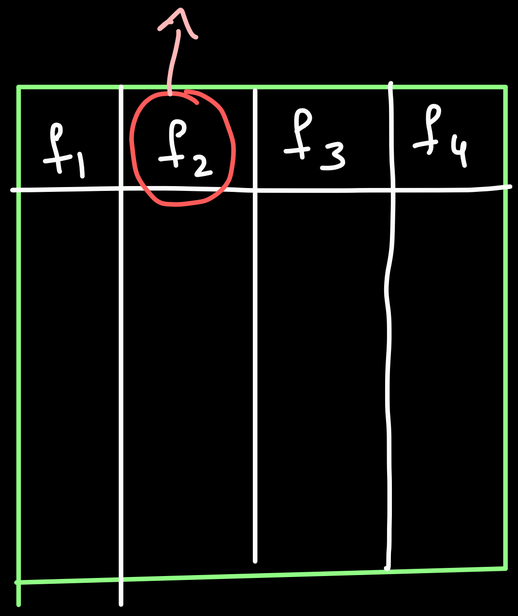


age = α

2024 = β

year = $\alpha = -1$

colinearity with multiple features ⇒ multicolinearity

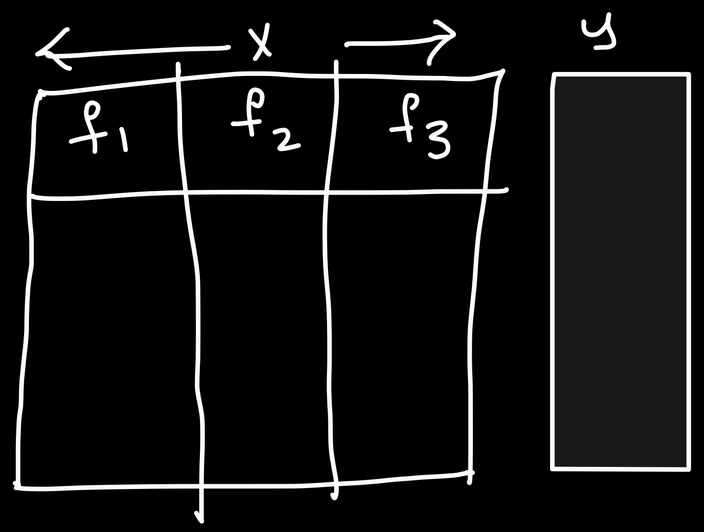


$$f_2 = \alpha_1 \cdot f_1 + \alpha_3 f_3 + \alpha_4 f_4 + \alpha_0$$

$$\Rightarrow F = \frac{9}{5} * c^{\circ} + 32$$

Height (cm)	X Height (ft)
⇒	

** why multicolinearity should not exists ?



$$w^* = [w_1, w_2, w_3] \quad w_0$$

$$= [1, 2, 3] \quad w_0 = 5 \Rightarrow f_3$$

$$x_q = [x_1 \; x_2 \; x_3]$$

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$

$$\hat{y} = 1 \cdot x_1 + 2 x_2 + 3 x_3 + 5 \quad \rightarrow (1)$$

$$x_2 = 1.5\, x_1$$

$$\hat{y} = 1 \cdot x_1 + 2(1.5\, x_1) + 3x_3 + 5$$

$$\hat{y} = 4x_1 + 3x_3 + 5 \longrightarrow [4, 0, 3] \quad 5$$

x_1

Eg:

$x_q = [2, 3, 1]$

$x_q \rightarrow [2, 3, 1]$

① $\hat{y} = 1 \cdot 2 + 2 * 3 + 3 * 1 + 5$

$= 16$ ✓

$\hat{y} = 4 * 2 + 3 * 1 + 5$

$= 16$ ✓

✓ problem : messed up weights ✓

✓ no feature importance

↳ unstability of weights ✓

VIF [Variance Inflation Fa

pearson =

How does multicollinearity affect regression analysis ?

4 users have participated

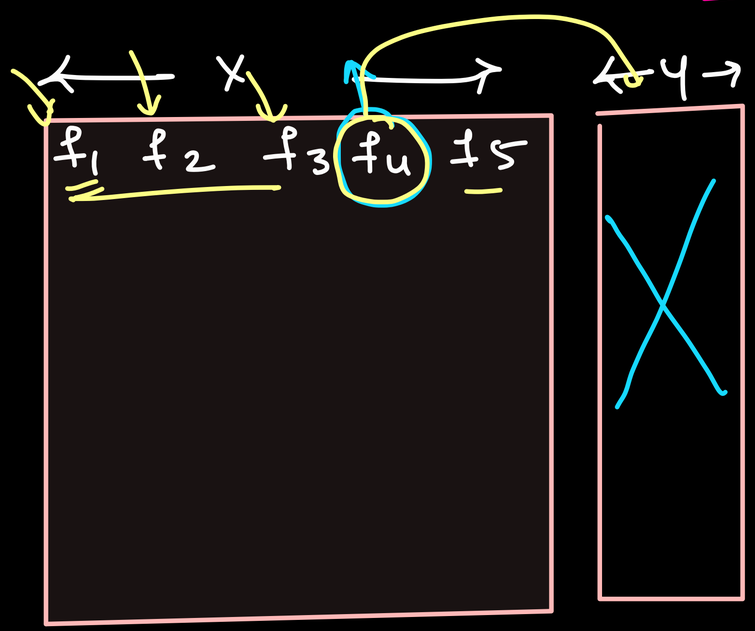
A	It reduces the interpretability of regression coefficients.	75%
B	It increases the accuracy of the regression model.	0%
C	It improves the goodness-of-fit of the regression model.	25%
D	It has no impact on the regression analysis.	0%

weights

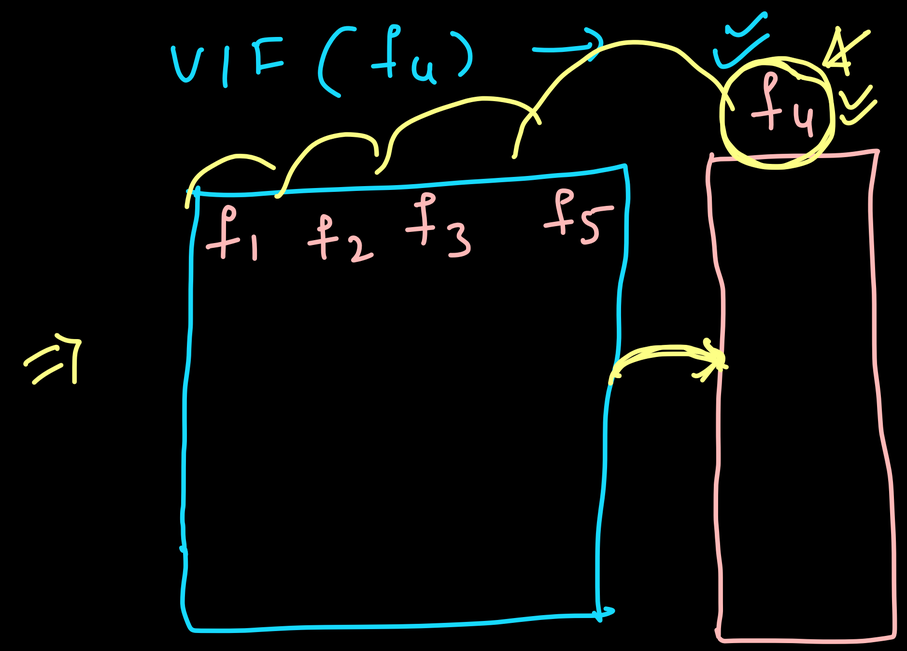
End Quiz Now

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VIF (Variance Inflation Factor)



{X, Y}



⇒ R2-score ↓

0.18

$$f_4 = w_1 f_1 + w_2 f_2 + w_3 f_3 + w_5 f_5 + w_0$$

$$VIF_j = \frac{1}{1 - R^2_j}$$

if $R2 \rightarrow 1$ $\Rightarrow$ $VIF \Rightarrow \infty$

Multi colinearity

if $R2 \rightarrow 0$ $\Rightarrow$ $VIF \Rightarrow 1$

Statsmodel

$VIF \rightarrow [1, \infty)$

No M.C

v.large M.C

$VIF(f_1) =$ ——

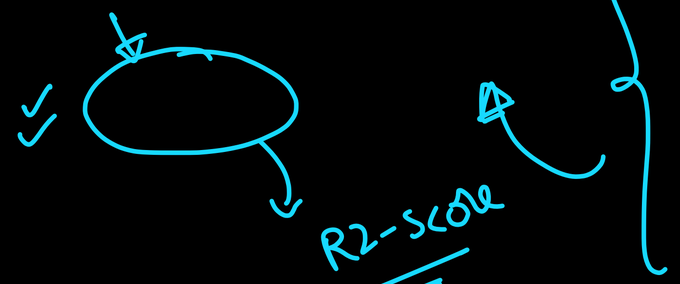
$VIF(f_2) =$ ——

⋮

$VIF(f_d) =$ ——

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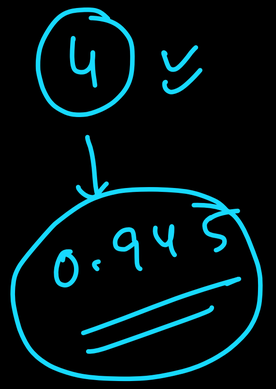
Thumb Rule



VIF > 10 : V. High M.C

VIF > 5 High M C

VIF < 5 low M C



⇒ NO Multi?

10 · 25 → 10 : 30

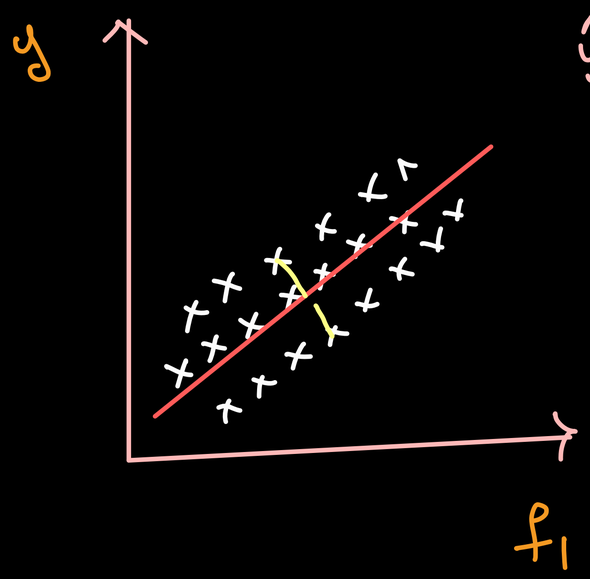
3. Normality of Residuals

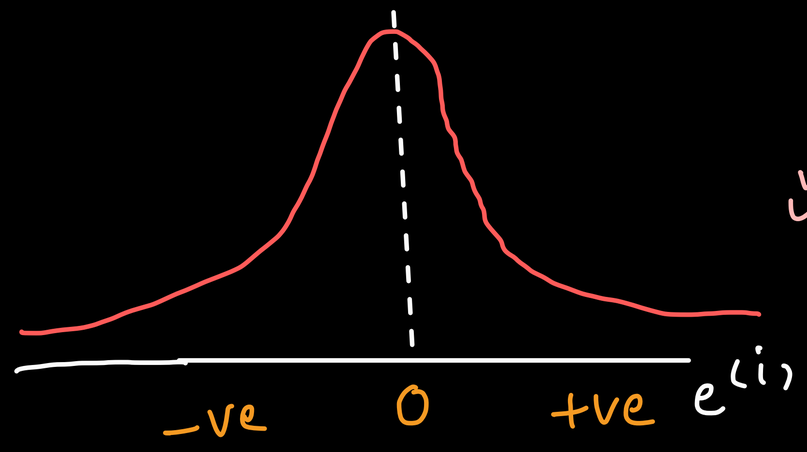
$y - \hat{y}$

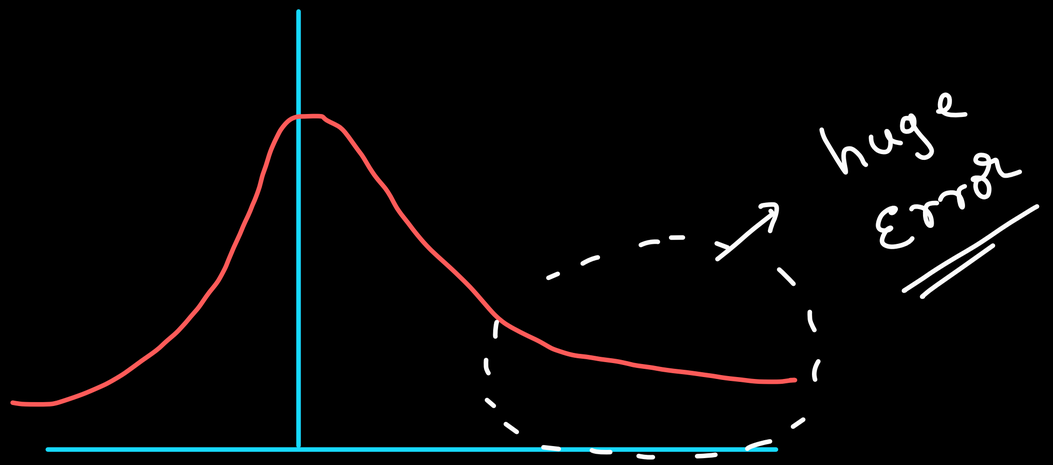
↳ residuals / errors should be normally distribution

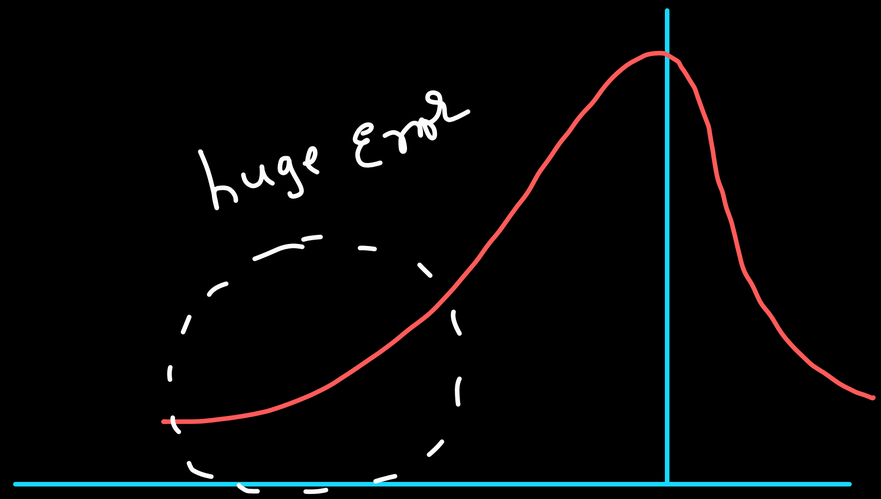
$$y = \underbrace{w^T \vec{x} + w_0}_{\hat{y}} + \epsilon \quad (\epsilon = \text{Error})$$

$$\epsilon = y - \hat{y} \rightarrow \text{residual}$$





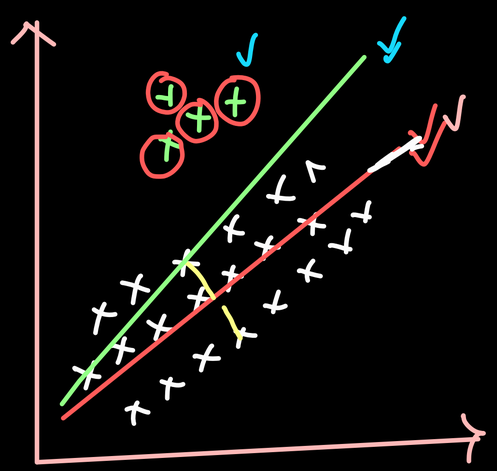






Impact of outliers in lin reg

Residual analysis



(1) detect the outlier → IQR, box plot

(2) Deal with outliers → remove & retrain the model

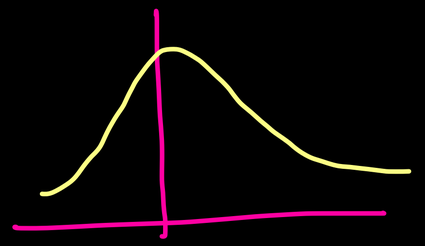
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While building a risk prediction model for loan defaulters, it was observed that the erros were right skewed.
Does this imply anyway that the linear regression model is inaccurate?

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A	Yes, since the features are multi-collinear	0%
✓ B	Yes, since the errors aren't normaly distributed	100%
C	Yes, by violation of assumption of linearity	0%
D	No, the model may be accurate.	0%

End Quiz Now

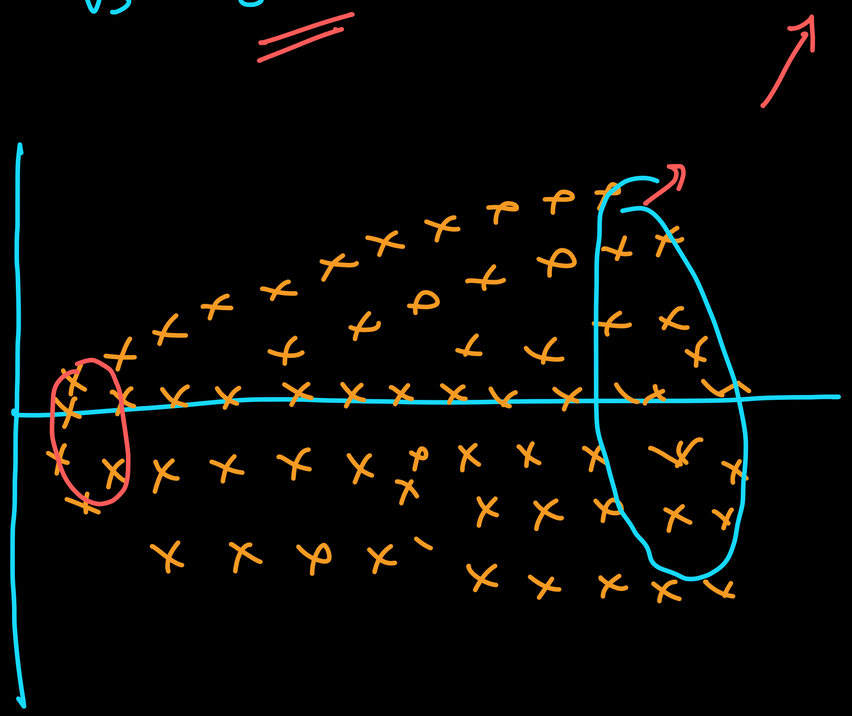


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4. No Heteroskedasticity

No Autocorrelation

$\Rightarrow e^{(i)}$ vs $y^{(i)}$



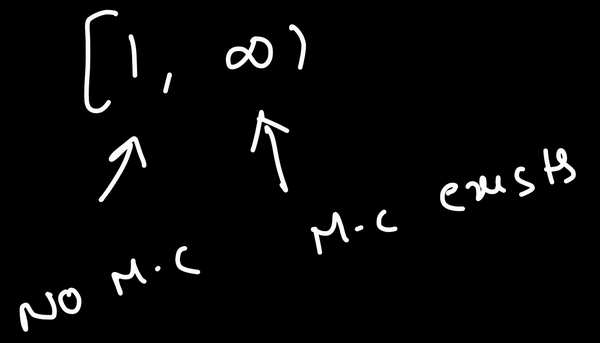
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In linear regression, a high VIF value suggests:

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A	Heteroskedasticity is present	0%
B	A strong linear relationship between the independent and dependent variables.	17%
C	The absence of outliers in the dataset.	0%
D	Strong multicollinearity between predictor variables.	83%

End Quiz Now



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